engineering 4 us all

Student Handout

Name: Sophia Vanta Tijam **Date:** 02/07 **Class:** 8

MyDesign Portfolio

Element B Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to structure your research findings around past and current attempts to solve the problem, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own research into prior design attempts, be sure to delete the instructions.

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Corsi-Rosenthal Cube>

https://encycla.com/Corsi-Rosenthal_Cube

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).


(https://encycla.com/Corsi-Rosenthal_Cube)

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Include at least one image of this prior design attempt. Make sure to cite all images.

Details: The Corsi-Rosenthal Cube or also known as the Comparetto Cube, is an air filter that you can make yourself. It costs around $100 and consists of a box fan and furnace filters. On Encycla, people say it works efficiently as a commercial HEPA air filter. (https://encycla.com/Corsi-Rosenthal_Cube)

Include main details about this prior design attempt

Analysis: This design is useful as it does help filter the air. But, this design doesn't look professional to put in at work or school. If it were to break or in need to replace the filters, it would be difficult to take apart and fix since it is taped together. If you were to want to put it in the middle of a classroom to clear the air, it can get in the way for students and teachers in the classroom.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #2 - <Ozone Generator>

Source:
https://ww2.arb.ca.gov/our-work/programs/air-cleaners-ozone-products/hazardous-ozone-generating-air-purifiers#:~:text=Indoor%20%22air%20purifiers%22%20or%20air,a%20healthy%20kind%20of%20oxygen.

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

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Image
(https://airthereal.com/products/ma10k-pro-ozone-generator)

Include at least one image of this prior design attempt. Make sure to cite all images.

Details: The ozone generator is an indoor air filter that purifies the air through ozone. Ozone is a colorless gas that is emitted into the atmosphere to clean the air. (https://ww2.arb.ca.gov/our-work/programs/air-cleaners-ozone-products/hazardous-ozone-generating-air-purifiers#:~:text=Indoor%20%22air%20purifiers%22%20or%20air,a%20healthy%20kind%20of%20oxygen.)

Include main details about this prior design attempt

Analysis: Most people don't use this generator because ozone is a toxic gas. Many manufacturers say that ozone is "activated oxygen" that "can purify the air and remove chemicals, mold, viruses, and bacteria." Ozone is a serious hazard and could cause serious health risks.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #3 - <Ionic Purifier>

Source:
https://www.epa.gov/indoor-air-quality-iaq/what-are-ionizers-and-other-ozone-generating-air-cleaners#:~:text=Ion%20generators%20act%20by%20charging,particles%20back%20to%20the%20unit.

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and

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research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image  *(https://www.bobvila.com/articles/best-ionic-air-purifiers/)*

Include at least one image of this prior design attempt. Make sure to cite all images.

Details: Ion filters remove smaller particles like tobacco smoke. They are either attached to the wall, floor, tabletops, or draperies. But it cannot attack stronger particles like gases and odors. They are ineffective against dust and pollen.

Include main details about this prior design attempt

Analysis: This filter is only able to get rid of small particles. This means that this won't be useful especially in the art wing. If it isn't able to get rid of smell, it can't be used in Mrs. Poole's room since the smell of resin is the main concern. It can't be used in Mrs. Hudson's room because projects are happening every second in that class and the ionic filter probably won't be able to filter it out.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #4 - <HEPA Filter>

Source:
https://www.filtrete.com/3M/en_US/filtrete/home-tips/full-story/~/how-to-reduce-dust-mites/?storyid=75200b03-7977-4621-a248-b5bc0d4e1ae2#:~:text=Standing%20for%20high%20efficiency%20particulate,in%20vacuums%20and%20air%20purifiers.

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Image 

(https://www.iqsdirectory.com/articles/air-filter/hepa-air-filters.html)

Include at least one image of this prior design attempt. Make sure to cite all images.

Details: This filter captures large particles like pollen, dust, smoke, and pet dander. The way it filters is through a fine mesh that puts these particles in. These fine meshes are normally in vacuums and other air purifiers. (https://www.filtrete.com/3M/en_US/filtrete/home-tips/full-story/~/how-to-reduce-dust-mites/?storyid=75200b03-7977-4621-a248-b5bc0d4e1ae2#:~:text=Standing%20for%20high%20efficiency%20particulate,in%20vacuums%20and%20air%20purifiers.)

Include main details about this prior design attempt

*Analysis***:** The HEPA filter is useful since it does trap odors and large particles. But the size of the HEPA filter is too big. It also might be too heavy to put into the ceiling.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #5 - <UV Purfier>

Source: https://molekule.com/blogs/all/how-do-uv-light-air-purifiers-work

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out

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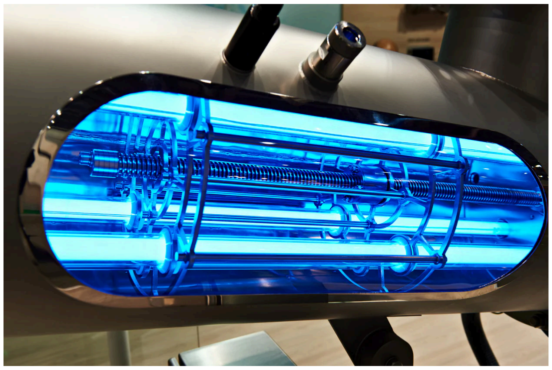
Image 
(https://molekule.com/blogs/all/how-do-uv-light-air-purifiers-work)

Include at least one image of this prior design attempt. Make sure to cite all images.

Details: UV stands for ultraviolet. This air filter uses UV lighting to clear the air. The short waves of light are able to eliminate microorganisms like mold, bacteria, and viruses. It works like other air purifiers. But it doesn't directly eliminate air pollutants. (https://molekule.com/blogs/all/how-do-uv-light-air-purifiers-work)

Include main details about this prior design attempt

*Analysis***:** This air filter is able to get rid of bacteria but it does not say if it can get rid of smell. Also the lighting can be distracting to others while they work. (Mrs. Poole normally turns off the lights for some classes to take photos, this ruins that.)

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.